FEATURE-BASED REGISTRATION OF DIGITAL ELEVATION DATA

Miguel Aguilera and A. Ben Hamza
Concordia Institute for Information Systems Engineering
Concordia University, Montréal, QC, Canada
e-mail: {jesus_ag, hamza}@encs.concordia.ca

Abstract

We present a feature-based registration method using Morse singularities and information theory. The main idea behind our proposed framework is to encode a 3D terrain digital elevation map into a set of Morse critical points. Then an entropic dissimilarity measure between the Morse features of the target and the reference maps is maximized to bring the elevation data into alignment. We also show that maximizing this divergence measure leads to minimizing the total length of the joint minimal spanning tree between features of the misaligned elevation maps. Illustrative experimental results clearly show the much improved performance and the georegistration accuracy of the proposed technique.

Keywords — *Geo-registration, Morse theory, minimum spanning tree, information theory.*

1 Introduction

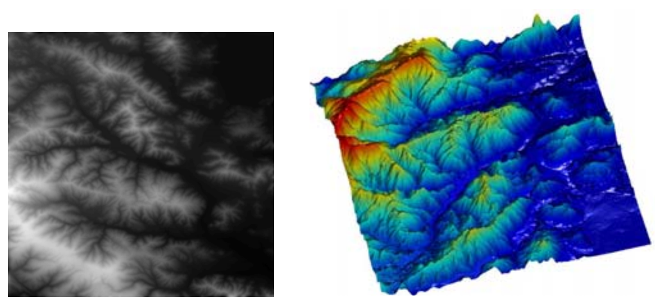
Image registration refers to the process of aligning images so that their details overlap accurately [1–3]. A wide range of registration techniques has been developed for many different types of applications and data, such as mean squared alignment, correlation registration, and moment invariant matching. Inspired by the successful application of the mutual information measure [1, 2], and looking to address its limitations in often difficult imagery, we recently proposed an information-theoretic approach to ISAR image registration using the Jensen-Rényi divergence [4]. This entropic measure enjoys appealing mathematical properties affording a great flexibility in a number of applications [5, 6].

Multisensor data fusion technology combines data and information from multiple sensors to achieve improved accuracies and better inference about the environment that could be achieved by the use of a single sensor alone. In this paper, we introduce a nonparametric multisensor data fusion algorithm for the registration of digital elevation maps (DEMs). The goal of georegistration is to align a target DEM to the geographic location of a reference DEM using global or feature-based techniques. Our proposed method falls into the category of feature-based techniques which require that features be extracted and described before two DEMs can be registered. The proposed approach consists of two major steps. The first step involves extracting Morse singularity features [7–9] from the DEMs represented as 3D surfaces. These geometric features form the so-called Reeb graph [8, 9] which is a topological representation of a digital elevation surface and has the advantage to be stored or transmitted with a much smaller amount of data. Inspired by the mutual information based approaches for image registration, we propose in the second step the use of an information-theoretic divergence based on Tsallis entropy [10] as a dissimilarity measure between Morse features of the target and reference DEMs. Tsallis entropy can be estimated using the length of the minimal spanning tree (MST) over Morse feature vectors of a DEM. Then, the registration is performed by minimizing the length of a joint MST which spans the graph generated from the overlapping target and reference DEMs [3].

The rest of the paper is organized as follows. The next section is devoted to the problem formulation. In Section 3, we describe the proposed method, and discuss in more detail its most important algorithmic steps. In Section 4, we provide experimental results to show the robustness and the georegistration accuracy of the proposed method. And finally, we conclude in Section 5.

2 Problem Formulation

A digital elevation map is a raster of elevation values, and consists of an array of points of elevations, sampled systematically at equally spaced intervals. We may represent a DEM as an image $I : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$ (see Figure 1(a)) where each image location denotes a height value.



(a) (b)

Figure 1. Illustration of a digital elevation map in 2D and 3D.

Let two digital elevation maps $I_1, I_2 : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$, where Ω is a bounded set (usually a rectangle). The goal of georegistration is to align the target I_1 to the geographic location of the reference I_2 by maximizing a dissimilarity measure D between I_1 and $\mathcal{T}_{(\boldsymbol{t},\theta,s)} I_2$, where $\mathcal{T}_{(\boldsymbol{t},\theta,s)}$ is a Euclidean transformation with translation parameter vector $\boldsymbol{t} = (t_x, t_y)$, a rotation parameter θ, and a scaling parameter s. In other words the registration problem may be succinctly stated as

$$(\boldsymbol{t}^\star, \theta^\star, s^\star) = \arg \max_{(\boldsymbol{t},\theta,s)} D(I_1, \mathcal{T}_{(\boldsymbol{t},\theta,s)} I_2),$$

or equivalently

$$(\boldsymbol{t}^\star, \theta^\star, s^\star) = \arg\max_{(\boldsymbol{t},\theta,s)} D(\mathcal{U}, \mathcal{T}_{(\boldsymbol{t},\theta,s)}\mathcal{V}),$$

where $\mathcal{U}$ and $\mathcal{V}$ are two sets of feature vectors extracted from I_1 and I_2 respectively. To georegister the target DEM to the reference DEM, we propose in this paper a feature-based approach by extracting Morse singular points from these DEMs, and then we use the Jensen-Tsallis divergence as a dissimilarity measure [10].

2.1 Morse singular points

A digital elevation map $I : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$ may be viewed as a 3D surface $\mathbb{M} \subseteq \mathbb{R}^3$ defined as $\mathbb{M} = \{(x, y, z) : z = I(x, y)\}$ where $z = I(x, y)$ is the height value on the DEM domain Ω (Figure 1(b)). Let $h : \mathbb{M} \to \mathbb{R}$ be a height function defined by $h(\boldsymbol{p}) = I(x, y)$ for all $\boldsymbol{p} = (x, y, I(x, y)) \in \mathbb{M}$, that is h is the orthogonal projection with respect to the z-axis. A point $\boldsymbol{p}_0$ on $\mathbb{M}$ is a singularity or critical point of h if $\boldsymbol{p}_0 = \boldsymbol{r}(x_0, y_0) = (x_0, y_0, I(x_0, y_0))$, for some $(x_0, y_0) \in \Omega$, and the gradient of $h \circ \boldsymbol{r}$ at (x_0, y_0) vanishes, i.e. $\nabla(h \circ \boldsymbol{r}(x_0, y_0)) = 0$. A singularity $\boldsymbol{p}_0$ is nondegenerate if the Hessian matrix $\nabla^2(h \circ \boldsymbol{r}(x_0, y_0))$ is nonsingular. We say that the height function is a Morse function if all its singular points are nondegenerate, that is minimum, maximum and saddle points.

2.2 Jensen-Tsallis Divergence

Let X be a continuous random variable with a probability density function f defined on $\mathbb{R}$. Shannon entropy is defined as $H(f) = -\int f(x) \log f(x)\, dx$, and it is a measure of uncertainty, dispersion, information, and randomness. The maximum uncertainty or equivalently minimum information is achieved by the uniform distribution. Hence, we can think of the entropy as a measure of uniformity of a probability distribution. A generalization of Shannon entropy is Tsallis entropy [11–13] given by

$$H_\alpha(f) = \frac{1}{1-\alpha}\Big(\int f(x)^\alpha\, dx - 1\Big),$$

and was first introduced by Havrda and Charvát in [11], who were primarily interested in providing another measure of entropy. Tsallis, however, appears to have been principally responsible for investigating and popularizing the widespread physics applications of this entropy which is referred to nowadays as Tsallis entropy [12].

Definition 1: Let $f_1, f_2, \ldots, f_n$ be n probability density functions. The Jensen-Tsallis divergence is defined as

$$D(f_1, \ldots, f_n) = H_\alpha\left(\sum_{i=1}^{n} \lambda_i f_i\right) - \sum_{i=1}^{n} \lambda_i H_\alpha(f_i),$$

where $H_\alpha(\cdot)$ is Tsallis entropy, and $\lambda = (\lambda_1, \ldots, \lambda_n)$ be a weight vector such that $\sum_{i=1}^{n} \lambda_i = 1$ and $\lambda_i \geq 0$.

Using the Jensen inequality, it is easy to check that the Jensen-Tsallis divergence is nonnegative for $\alpha > 0$. It is also symmetric and vanishes if and only if the probability density functions $f_1, f_2, \ldots, f_n$ are equal, for all $\alpha > 0$.

Unlike other entropy-based divergence measures such as the Kullback-Leibler divergence, the Jensen-Tsallis divergence has the advantage of being symmetric and generalizable to any arbitrary number of probability density functions or data sets, with a possibility of assigning weights to these density functions. The following result establishes the convexity of the Jensen-Tsallis divergence of a set of probability density functions [13].

Proposition 1: For $\alpha \in [1, 2]$, the Jensen-Tsallis divergence is a convex function of $f_1, f_2, \ldots, f_n$.

In the sequel, we will restrict $\alpha \in [1, 2]$, unless specified otherwise. In addition to its convexity property, the Jensen-Tsallis divergence is an adapted measure of disparity among n probability density functions [10].

2.3 Minimal Spanning Tree

Let $\mathcal{V} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ be a set of feature vectors (e.g. Morse singularities), where $\boldsymbol{v}_i \in \mathbb{R}^d$ (e.g. $d = 3$ for 3D data). A spanning tree $\mathcal{E}$ is a connected acyclic graph that passes through all features and it is specified by an ordered list of edges e_{ij} connecting certain pairs $(\boldsymbol{v}_i, \boldsymbol{v}_j), i \neq j$, along with a list of edge adjacency relations. The edges e_{ij} connect all n features such that there are no paths in the graph that lead back to any given feature vector. The total length $L_{\mathcal{E}}(\mathcal{V})$ of a tree is given by

$$L_{\mathcal{E}}(\mathcal{V}) = \sum_{e_{ij} \in \mathcal{E}} \|e_{ij}\|.$$

The minimal spanning tree $\mathcal{E}^\star$ is the spanning tree that minimizes the total edge length $L_{\mathcal{E}}(\mathcal{V})$ among all possible spanning trees over the given features

$$L^\star(\mathcal{V}) = \sum_{e_{ij} \in \mathcal{E}^\star} \|e_{ij}\| = \min_{\mathcal{E}} L_{\mathcal{E}}(\mathcal{V}).$$

The set $\mathcal{V}$ is called a random feature set if its elements are random variables with a probability density function f. Then it can be shown that

$$\lim_{n\to\infty} \frac{L^\star(\mathcal{V})}{n^\alpha} = \beta \int f(x)^\alpha d\,x \qquad a.s. \tag{1}$$

where the constant β plays a role of bias correction [3]. Hence, we may define an estimator $\widehat{H}_\alpha$ of Tsallis entropy as

$$\widehat{H}_\alpha(\mathcal{V}) = \frac{1}{1-\alpha}\left[\frac{L^\star(\mathcal{V})}{\beta\, n^\alpha} - 1\right], \tag{2}$$

with $L^\star(\mathcal{V})$ the total length of the MST. This estimator is asymptotically unbiased and almost surely consistent estimator of the Tsallis entropy [3].

3 Proposed approach

The goal of our proposed approach may be described as follows: Given two DEMs to be registered, we first extract their Morse singularity features, and we compute a joint MST connecting their Morse features, then we optimize the Jensen-Tsallis divergence to bring these DEMs into alignment. Without loss of generality, we consider the transformation $\mathcal{T}_{\boldsymbol{\ell}}$, where $\boldsymbol{\ell} = (\boldsymbol{t}, \theta)$, that is a Euclidean transformation with translation parameter vector $\boldsymbol{t} = (t_x, t_y)$, and a rotation parameter θ. In other words, for $\boldsymbol{x} = (x, y)$ we have $\mathcal{T}_{\boldsymbol{\ell}}(\boldsymbol{x}) = R\boldsymbol{x} + \boldsymbol{t}$, where R is a rotation matrix given by

$$R = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

Let I_1 and I_2 be two DEMs to be registered. The proposed methodology may now be described as follows:
(i) Find the Morse singularity features $\mathcal{U} = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\}$ and $\mathcal{V} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$ of I_1 and I_2 respectively.
(ii) Transform $\mathcal{V}$ to a new set $\mathcal{W} = \mathcal{T}_{\boldsymbol{\ell}}\mathcal{V} = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_m\}$
(iii) Find the optimal parameter vector $\boldsymbol{\ell}^\star = (\boldsymbol{t}^\star, \theta)$ of the Jensen-Tsallis divergence

$$\boldsymbol{\ell}^\star = \arg\max_{\boldsymbol{\ell}} D(\mathcal{U}, \mathcal{W}), \tag{3}$$

where

$$D(\mathcal{U}, \mathcal{W}) = \widehat{H}_\alpha(\mathcal{U} \cup \mathcal{W}) - \lambda \widehat{H}_\alpha(\mathcal{U}) - (1-\lambda)\widehat{H}_\alpha(\mathcal{W}),$$

and $\lambda = k/(k+m)$.

3.1 Extraction of Morse singularities

We applied the algorithm introduced by Takahashi *et al.* [9] to extract Morse singular points from the target and reference DEMs. The algorithm extracts the singularity points while preserving the topological integrity of the surface of a DEM. This means that the extracted singular points must satisfy the Euler-Poincaré formula given by

$$\chi = \#\text{minima} - \#\text{saddlepoints} + \#\text{maxima} = 2.$$

3.2 Optimization of the dissimilarity measure

Recall from Proposition 1 that the Jensen-Tsallis divergence is convex when $\alpha \in [1, 2]$. Using and Eq. (2) and Eq. (3), it is clear that solving

$$\boldsymbol{\ell}^\star = \arg\max_{\boldsymbol{\ell}} D(\mathcal{U}, \mathcal{T}_{\boldsymbol{\ell}}\mathcal{V}),$$

may be reduced to solving

$$\begin{aligned} \boldsymbol{\ell}^\star &= \arg\max_{\boldsymbol{\ell}} \widehat{H}_\alpha(\mathcal{U} \cup \mathcal{T}_{\boldsymbol{\ell}}\mathcal{V}) \\ &= \arg\max_{\boldsymbol{\ell}} \frac{1}{1-\alpha}\left[\frac{L^\star(\mathcal{U} \cup \mathcal{T}_{\boldsymbol{\ell}}\mathcal{V})}{\beta\,(k+m)^\alpha} - 1\right], \end{aligned}$$

or equivalently solving the minimization problem

$$\boldsymbol{\ell}^\star = \arg\min_{\boldsymbol{\ell}} L^\star(\mathcal{U} \cup \mathcal{T}_{\boldsymbol{\ell}}\mathcal{V}) = \arg\min_{\boldsymbol{\ell}} \sum_{e_{ij} \in \mathcal{E}^\star} \|e_{ij}\|, \tag{4}$$

where $\mathcal{E}^\star$ is the MST of $\mathcal{U} \cup \mathcal{T}_{\boldsymbol{\ell}}\mathcal{V}$, and $\|e_{ij}\|$ is the edge length which depends on the transformation parameter $\boldsymbol{\ell}$ and is given by

$$\begin{aligned} \|e_{ij}\| &= \sqrt{(I_1(\boldsymbol{x}_i) - I_1(\boldsymbol{x}_j))^2 + (\mathcal{T}_{\boldsymbol{\ell}} I_2(\boldsymbol{x}_i) - \mathcal{T}_{\boldsymbol{\ell}} I_2(\boldsymbol{x}_j))^2} \\ &= \sqrt{(I_1(\boldsymbol{x}_i) - I_1(\boldsymbol{x}_j))^2 + (I_2(R\boldsymbol{x}_i + \boldsymbol{t}) - I_2(R\boldsymbol{x}_j + \boldsymbol{t}))^2}, \end{aligned}$$

where $\boldsymbol{x}_i = (x_i, y_i) \in \Omega$, and $\boldsymbol{x}_j = (x_j, y_j) \in \Omega$.

Hence, the DEM registration is performed by minimizing the total length of the minimum spanning tree which spans the joint MST generated from the overlapping feature vectors of the target and the reference DEMs. The minimization problem given by Eq. (4) may be solved using the method of steepest descent [14]. Let $\boldsymbol{\ell}^r$ be the value of the objective function given by Eq. (4) at the r-th iteration. At each iteration, the update rule is given by

$$\boldsymbol{\ell}^{r+1} = \boldsymbol{\ell}^r + \alpha\,\boldsymbol{d}^r = \boldsymbol{\ell}^r - \alpha \sum_{e_{ij} \in \mathcal{E}^\star} \nabla\|e_{ij}\|,$$

where $\boldsymbol{d}^r = -\sum_{e_{ij} \in \mathcal{E}^\star} \nabla\|e_{ij}\|$ is the direction vector of steepest descent that is the direction for which the objective function will decrease the fastest, and α is a scalar which determines the step size taken in that direction. Taking the gradient of the edge length with respect to $\boldsymbol{\ell}$ yields

$$\begin{aligned} \nabla\|e_{ij}\| &= \frac{1}{\|e_{ij}\|}\Big(I_2(R\boldsymbol{x}_i + \boldsymbol{t}) - I_2(R\boldsymbol{x}_j + \boldsymbol{t})\Big) \\ &\times \Big(\nabla I_2(R\boldsymbol{x}_i + \boldsymbol{t}) \cdot J_{R\boldsymbol{x}_i + \boldsymbol{t}} - \nabla I_2(R\boldsymbol{x}_j + \boldsymbol{t}) \cdot J_{R\boldsymbol{x}_j + \boldsymbol{t}}\Big), \end{aligned}$$

where $J_{R\boldsymbol{x}+\boldsymbol{t}}$ is the Jacobian matrix given by

$$J_{R\boldsymbol{x}+\boldsymbol{t}} = \begin{pmatrix} 1 & 0 & -x\sin\theta + y\cos\theta \\ 0 & 1 & -x\cos\theta - y\sin\theta \end{pmatrix}.$$

4 Experimental Results

In this section we present some experimental results to show the much improved performance of the proposed method in DEM registration. We applied a Euclidean transformation to the reference DEM, with known parameters (t_x, t_y, θ). To georegister these two DEMs, we then applied the iterative gradient descent algorithm described in Section 3 to find the optimal parameters $t_x^\star$, $t_x^\star$ and $\theta^\star$. The registration results are provided in Table I, where the errors between the original and the estimated transformation parameters are also listed. The estimated values clearly indicate the effectiveness and the accuracy of the proposed algorithm in georegistering elevation data. The joint MST of the target and reference is displayed in Figure 2, and also in Figure 3 where the overlapping DEM surfaces are shown as well. Note that for ease of visualization, only the maxima features are displayed.

5 Conclusions

We proposed a graph-theoretic technique for registration of digital elevation maps using Morse singularities and

$\boldsymbol{\ell}$			$\boldsymbol{\ell}^\star$			**error** $= \boldsymbol{\ell} - \boldsymbol{\ell}^\star$		
t_x	t_y	θ	$t_x^\star$	$t_y^\star$	$\theta^\star$	t_x^e	t_y^e	θ^e
2	4	0	2.64	3.20	1.26	0.64	0.80	1.26
2	4	5	2.02	3.65	5.70	0.02	0.35	0.70
5	10	15	5.77	9.53	15.34	0.77	0.47	0.34
4	8	20	3.46	7.38	20.87	0.54	0.62	0.87

TABLE I
Registration results.

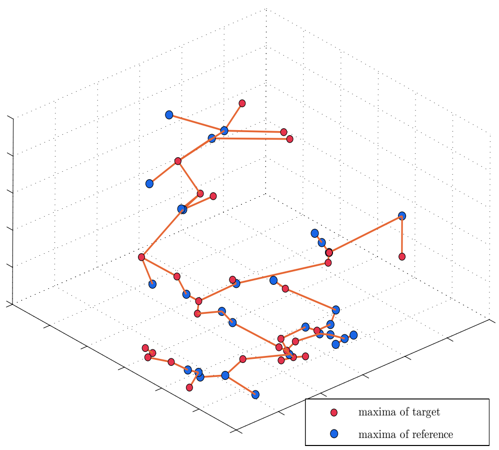


Figure 2. Joint MST of target and reference DEMs (only maxima features are displayed).

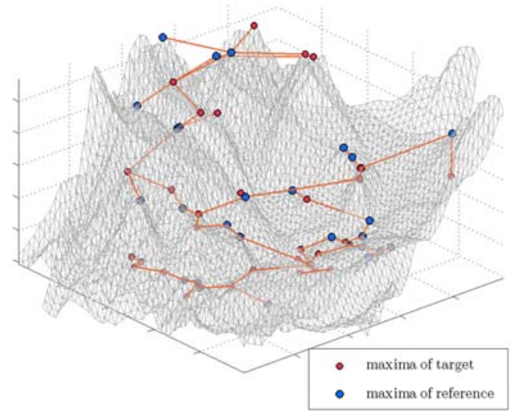


Figure 3. Overlapping surfaces and joint MST of target and reference DEMs (only maxima features are displayed).

an entropic dissimilarity measure. The georegistration is achieved by minimizing the total length of the joint minimal spanning tree of the target and the reference elevation maps. The main advantages of the proposed approach are: (i) Tsallis entropy provides a reliable data estimator, (ii) the proposed approach is simple and computationally fast, and (iii) the experimental results provide accurate registration results and clearly indicate the suitability of the proposed method for georegistration of 3D terrain elevation data.

Acknowledgments

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